1 Numerical Experiments

All experiments use $h = J = -1.0$, $\delta t = 0.1$, $\epsilon_{\text{cut}} = 10^{-11}$, and float32 precision on AMD MI100 (GFX908, 32 GB HBM2), unless MI60 (GFX906) is explicitly noted. The center observable is Z_c at site $(\lfloor (N_x - 1)/2 \rfloor, \lfloor (N_y - 1)/2 \rfloor)$, the geometric center of the grid.

The primary observable throughout is the *spin autocorrelation function*

$$C(t) = \frac{1}{2^N} \operatorname{Tr}\big[Z_c(t)\, Z_c\big], \qquad Z_c(t) = e^{i\mathcal{H}t}\, Z_c\, e^{-i\mathcal{H}t}, \tag{1}$$

where N is the number of sites and Z_c is the Pauli-Z operator at the lattice center. $C(t)$ measures the overlap between the Heisenberg-evolved operator $Z_c(t)$ and its initial form Z_c: it equals 1 at $t = 0$ by normalization, and decreases toward 0 as $Z_c(t)$ scrambles into many-body Pauli strings that are orthogonal to Z_c. This decay is purely *unitary* — there is no dissipation or energy loss; $C(t) \to 0$ means the operator has spread across the entire lattice, not that it has decayed.

1.1 Correctness Validation

The implementation was validated against:

1. **Julia reference**: $C(t)$ for 4×4 at $\chi = 50$ matches the Julia float64 implementation to within float32 rounding ($< 10^{-6}$ absolute error) for $t \leq 1.6$.
2. **Chi consistency**: for $t \leq 1.4$, $C(t)$ agrees across $\chi \in \{40, 50, 60, 80, 110\}$ to 5 significant figures (Table 1), confirming that truncation error is negligible for early-to-mid time evolution at these bond dimensions.

1.2 χ-Convergence of $C(t)$ on 4×4 (MI100)

Table 1 reports $C(t)$ at selected Trotter layers for $\chi \in \{20, 40, 50, 60, 70, 80, 100, 110\}$ on a 4×4 grid. Bold entries are BP metastable traps.

Table 1 reveals the full structure of χ-convergence:

Early time ($t \leq 1.4$): universal agreement. All eight χ values agree to 5 significant figures through $t = 1.4$, confirming that bond dimension plays no role until entanglement saturates the smallest bond.

Monotone convergence for clean runs. Ignoring trapped layers (§1.3), the clean runs form a strictly decreasing sequence in χ at every late layer. At $t = 2.0$:

$$C(t)(\chi = 40) > C(t)(\chi = 50) > C(t)(\chi = 60) > C(t)(\chi = 80) > C(t)(\chi = 110), \tag{2}$$

with values $0.0572 > 0.0558 > 0.0551 > 0.0543 > 0.0537$. This monotonicity is not a coincidence: larger χ spans a strictly larger variational manifold, so the Bethe approximation is tighter from above. The gap between successive levels shrinks rapidly: from $\chi = 40$ to $\chi = 60$ the drop is 0.0021; from $\chi = 60$ to $\chi = 110$ it is only 0.0014 across 50 more bond dimension units. We identify $\chi^* \approx 60$ as the point of diminishing returns for this system: below it, each increment in χ buys meaningful accuracy; above it, the gain is less than 0.2% per step and the computational cost grows as χ^4.

Table 1: $C(t)$ on 4×4, MI100, for $\chi \in \{20, \ldots, 110\}$. **Bold**: metastable BP trap ($C(t) \approx 0.007$–0.009), identified by elevated sweep count and inconsistency with neighboring layers. †: $\chi = 20$ at $t = 2.0$ is dominated by truncation error (gate_err $\approx 5 \times 10^{-4}$); values above $t = 1.5$ are an overestimate of the true $C(t)$. Clean runs form a monotone converging sequence in χ at every layer.

t	$\chi = 20$	$\chi = 40$	$\chi = 50$	$\chi = 60$	$\chi = 70$	$\chi = 80$	$\chi = 100$	$\chi = 110$
0.1	0.922619	0.922619	0.922619	0.922619	0.922619	0.922619	0.922619	0.922619
0.5	0.175284	0.175284	0.175284	0.175284	0.175284	0.175284	0.175284	0.175284
0.7	0.128375	0.128376	0.128376	0.128376	0.128376	0.128376	0.128376	0.128376
0.8	0.134860	0.134861	0.134860	0.134860	0.134859	0.134859	0.134859	0.134859
1.0	0.126956	0.126956	0.126956	0.126957	0.126956	0.126956	0.126956	0.126956
1.2	0.103570	0.103570	0.103552	0.103553	0.103551	0.103555	0.103555	0.103554
1.4	0.085268	0.085328	0.085247	0.085250	0.085239	0.085262	0.085256	0.085257
1.6	0.071531	0.071505	0.071163	0.071154	0.071198	0.071167	0.071106	0.071107
1.7	0.066268	0.066207	0.065696	0.065558	0.065609	**0.008782**	0.065395	**0.008772**
1.8	0.062166	0.062173	0.061469	0.061110	0.061130	0.060939	0.060710	0.060703
1.9	0.059252	0.059274	0.058282	0.057719	0.057660	0.057281	0.056919	0.056893
2.0	0.063100†	0.057169	0.055800	0.055145	**0.007200**	0.054300	**0.007100**	0.053695

† $\chi = 20$ at $t = 2.0$: truncation-limited; gate_err $\approx 5 \times 10^{-4}$, overestimates true $C(t)$.

Trapped layers. Four entries deviate sharply from the converging sequence: $\chi = 80$ and $\chi = 110$ at $t = 1.7$ ($C(t) \approx 0.009$), and $\chi = 70$ and $\chi = 100$ at $t = 2.0$ ($C(t) \approx 0.007$). All four recover in the subsequent layer (or, in the last-layer cases, cannot recover). These are BP metastable traps, analyzed in §1.3. Their irregular distribution in χ — appearing at 70 and 100 but not at 60, 80, or 110 — confirms that they are a property of the BP fixed-point landscape at specific (χ, t) combinations, not a monotone failure mode.

1.3 BP Fixed-Point Multiplicity and Trap Remediation

Four entries in Table 1 deviate sharply from the converging sequence: $\chi \in \{80, 110\}$ at $t = 1.7$ and $\chi \in \{70, 100\}$ at $t = 2.0$, all collapsing to $C(t) \approx 0.007$–0.009 before recovering in the next layer (or, in the last-layer cases, ending the run in the wrong state). These are *metastable BP traps*: BP has converged, in a finite number of sweeps, to a valid but physically wrong fixed point of the Bethe equations.

Two failure modes. Across the full (χ, t) parameter space we identify two qualitatively distinct BP failure modes that must not be confused.

Type 1 — metastable trap: BP converges in a finite number of sweeps (delta $< \epsilon_{\text{tol}} = 10^{-5}$), yet $C(t)$ collapses to ≈ 0.007–0.010 regardless of χ, inconsistent with all neighboring layers. The sweep count at the trapped layer is 2–3× the typical value (e.g., 49 sweeps at $\chi = 110$, $t = 1.7$ vs. 12–18 on neighboring layers), providing an early-warning signal that can be checked at runtime.

Type 2 — non-convergence: BP exhausts the 100-sweep budget with delta $\gg \epsilon_{\text{tol}}$. The resulting $C(t)$ is unreliable, but the next Trotter layer typically recovers; the event is self-announcing (sweep count = 100 and large delta). Crucially, Type 2 does not imply a wrong answer: MI60 $\chi = 50$, $t = 1.9$ fails to converge (100 sweeps, delta $= 1.80 \times 10^{-5}$) yet produces $C(t) = 0.060$, agreeing with the MI100 clean run to within 3%.

The sweep count alone does not distinguish the types; the final delta is essential. A runtime

diagnostic should flag any layer where (sweeps ≥ 30) *and* (delta $< \epsilon_{\text{tol}}$) as a candidate Type 1 trap, warranting a retry.

Fixed-point multiplicity. The Bethe free energy on a loopy lattice generically has multiple local minima. BP finds one depending on the message trajectory, which is determined by the update rule and the floating-point arithmetic path. Two controlled experiments provide direct evidence of competing fixed points.

Cross-architecture experiment. The trap patterns in Table 1 are *complementary* across architectures: MI100 traps at ($\chi = 80$, $t = 1.7$) and ($\chi = 70$, $t = 2.0$) while MI60 is clean at those points; conversely, MI60 traps at ($\chi = 50$, $t = 2.0$) and ($\chi = 80$, $t = 2.0$) while MI100 is clean. Both GPUs execute the same algorithm in float32, but MFMA (MI100) and scalar vector-ALU (MI60) reduction trees differ in rounding order, perturbing the message trajectory enough to select a different basin. When both are un-trapped, $C(t)$ agrees to within 1–2×10^{-3}.

Update-rule experiment. Fixing $\chi = 80$ on MI100, we compare undamped BP ($\alpha = 1$) and damped BP with $\alpha \in \{0.50, 0.80\}$. Damping introduces a momentum term into the message update:

$$\mu_e^{(k+1)} \leftarrow (1-\alpha)\,\mu_e^{(k)} + \alpha\,\hat{\mu}_e^{(k+1)}, \tag{3}$$

where $\hat{\mu}_e^{(k+1)}$ is the undamped update. Spectrally, damping maps an eigenvalue λ of the BP Jacobian to $(1-\alpha) + \alpha\lambda$, compressing the spectrum toward 1 and suppressing oscillatory modes that otherwise overshoot the fixed point. The result at ($\chi = 80, t = 1.7$): $\alpha = 0.50$ avoids the trap entirely ($C(t) = 0.066$), while $\alpha = 0.80$ and undamped both find the wrong basin ($C(t) = 0.009$, identical to six decimal places). At ($\chi = 80, t = 1.9$): the roles reverse — $\alpha = 0.80$ finds the correct fixed point ($C(t) = 0.057$) while $\alpha = 0.50$ does not converge.

Trap remediation. A key practical conclusion is that traps are *not fatal*: the correct $C(t)$ can be recovered by retrying the BP step with a different α. This is confirmed by the cross-α experiment: MI100 damped $\alpha = 0.50$ at ($\chi = 80, t = 1.7$) recovers $C(t) = 0.066$, matching MI60 undamped ($C(t) = 0.066$) to within 1.5×10^{-3} — two different hardware paths arriving at the same physical fixed point. The procedure is:

1. After each BP run, check whether (sweeps ≥ 30) and (delta $< \epsilon_{\text{tol}}$). If so, flag as candidate Type 1 trap.
2. Verify by comparing $C(t)$ to the previous layer: a drop of $> 50\%$ with no corresponding increase in gate_err is a strong signal.
3. Retry BP with a different α (e.g., try $\alpha = 0.50$ if undamped trapped; try $\alpha = 0.80$ if $\alpha = 0.50$ trapped).
4. If all α values fail simultaneously, the bond dimension is insufficient: increase χ rather than tuning α.

Capacity boundary. A distinct third regime appears at high entanglement saturation ($S_e / \log \chi \gtrsim 0.50$): *both* α values fail simultaneously. At ($\chi = 90$, $t = 1.9$) and ($\chi = 100$, $t = 2.0$), no choice of α recovers the correct $C(t)$ — the bond dimension itself is the bottleneck. This boundary is cleanly separated from the fixed-point multiplicity regime: below it, at least one α recovers the correct $C(t)$; above it, none does. The appropriate response is to increase χ, not to tune the update rule.

The new MI60 extended runs complete the portability picture at the portable operating points. At $\chi = 40$ and $\chi = 60$, both MI100 and MI60 are trap-free and agree to within 5×10^{-4} ($C(t)(\chi = 60, t = 2.0)$: MI100 = 0.0551, MI60 = 0.0546). These are the safe operating points for this system: the BP landscape is simple enough that neither damping nor architecture choice matters. Above $\chi \approx 60$, competing fixed points emerge and hardware-dependent trap patterns appear; below $\chi \approx 40$, truncation error dominates.

1.4 Operator Spreading and Lightcone Saturation

The spin autocorrelation $C(t) = \mathrm{Tr}[Z_c(t)\, Z_c]/2^N$ measures how much the evolved operator $Z_c(t)$ still overlaps its initial form Z_c at the center site. Under unitary evolution, $C(t) \to 0$ does *not* signal dissipation: there is no energy loss, no bath. Instead it signals *operator scrambling*: $Z_c(t)$ has spread across the lattice into many-body Pauli strings that are orthogonal to the initial local Z_c. The rate and geometry of this spreading are constrained by a Lieb–Robinson bound [1]: at coupling $|J| = 1$, the operator front propagates at speed $v_{\mathrm{LR}} \leq 2|J|$, reaching the lattice corner at Manhattan distance $d_{\max}$ after at least

$$n^* = \frac{d_{\max}}{|J|\,\delta t} = 10\, d_{\max} \quad \text{Trotter layers.} \tag{4}$$

For $n < n^*$, the boundary has not yet been reached and $C(t)$ probes bulk scrambling; for $n \geq n^*$, finite-size effects enter and $C(t)$ approaches its long-time plateau. Table 2 lists n^* for each grid.

Table 2: Lightcone saturation: grid size, geometric center, maximum Manhattan distance $d_{\max}$ from center to corner, and minimum steps n^* to reach the boundary. Runs with $n < n^*$ observe bulk scrambling only.

Grid	Center	$d_{\max}$	n^* (layers)	Layers run
3×3	$(1, 1)$	2	20	20
4×4	$(1, 1)$	4	20	20
5×5	$(2, 2)$	4	20	30
6×6	$(2, 2)$	6	30	30
7×7	$(3, 3)$	6	30	40
8×8	$(3, 3)$	8	40	40
9×9	$(4, 4)$	8	40	50
10×10	$(4, 4)$	9	45	50

Figure **??** (GIF / static strip) shows the per-site non-identity Pauli weight $n(r, t)$ on a 4×4 grid at $\chi = 50$ as a function of time. At $t = 0$, the operator is localized: $n(r, 0) = 1$ at the center site, $n(r, 0) = 0$ elsewhere. By $t = 0.5$ (n^* layers for this grid), weight has reached the boundary and the pattern has approximately C_4 symmetry imposed by the lattice. Corner sites accumulate the highest weight at late times ($n \approx 0.13$), consistent with the operator having spread maximally given the open boundary conditions.

Table 3 compares $C(t)$ trajectories across grid sizes at $\chi = 50$ (MI100), demonstrating both the common early-time behavior and the grid-dependent saturation.

Three regimes are visible in Table 3:

1. **Early-time universality ($t \leq 0.9$)**: all grids agree, because the operator front has not yet reached any boundary regardless of grid size.

Table 3: $C(t)$ at selected Trotter layers for $n \times n$ grids, $\chi = 50$, MI100. All grids share the same $C(t)$ for $t \leq 0.9$ to 4 significant figures (early-time universality). Divergence begins when the operator front reaches the lattice boundary, consistent with the lightcone bound n^* in Table 2.

t	3×3	4×4	5×5	6×6	7×7	8×8	9×9	10×10
0.9	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
1.4	0.113	0.085	0.096	0.096	0.096	0.096	0.096	0.096
1.9	0.101	0.058	0.060	0.060	0.060	0.065	0.065	0.065
2.4	—	—	0.050	0.050	—	—	—	—
2.9	—	—	0.038	0.038	0.039	0.020	0.020	0.019
3.9	—	—	—	—	0.002	0.000	0.000	0.000

— : beyond run length or beyond saturation.

2. **Boundary entry ($n^* \leq n \leq 2n^*$)**: $C(t)$ begins to depend on grid size as the front reflects from the boundaries. The 3×3 grid shows the strongest finite-size effects ($C(t)$ rises at $t \approx 0.9$ as the operator re-focuses after reflection).
3. **Scrambling plateau ($n \gg n^*$)**: $C(t) \to 0$ as the operator distributes across all Pauli strings. The rate is set by the system size: larger grids support more scrambled configurations, so $C(t)$ decays more slowly.

Odd-grid center. A subtle correctness issue affects odd-dimension grids. The center site must be the *geometric* center, $(N_x - 1)/2, (N_y - 1)/2)$ (integer division), to ensure that the lightcone is symmetric: any asymmetry in the initial condition creates an asymmetric entanglement structure that is harder for the Bethe approximation to maintain. Using the off-center formula $N_x/2 - 1$ (valid only for even grids) displaces the center by one site on 5×5 ($\Delta = 1$) and 7×7 ($\Delta = 1$) grids, producing an asymmetric $n(r, t)$ pattern and causing the Bethe norm to exceed 2.0 prematurely. With the corrected center, the 9×9 grid at $\chi = 20$ runs 50 layers with maximum Bethe norm 1.005 — the symmetric initial condition produces an entanglement structure that the Bethe approximation handles cleanly. With the wrong center, the same run explodes to norm 5.09 at layer 21.

1.5 Finite-Size Scaling and Bethe Approximation Stability

As the grid grows, two effects compound: more closed loops in the lattice make the Bethe approximation less accurate (increasing the effective χ^* needed for a given gate error), and the larger variational space requires more BP sweeps to converge. We characterize both effects using the Bethe norm $\|\psi\|^2_{\text{Bethe}} = \exp(\sum_v \log f_v - \sum_e \log Z_e)$, which equals 1 under the exact Bethe approximation and deviates when the approximation breaks down.

The gate error at saturation grows steadily with grid size at fixed χ, reflecting the increasing difficulty of maintaining the Bethe approximation on more densely connected lattices. For 5×5 and 6×6 at $\chi = 50$, gate_err is marginal (~ 2–7×10^{-3}) but $C(t)$ plateaus smoothly, indicating that the truncation error has not corrupted the correlation function. By 7×7, $\chi = 50$ is insufficient (gate_err $> 10^{-2}$) and $\chi = 60$ is needed; at 8×8 and beyond, $\chi = 20$ is the feasibility ceiling on MI100 for multi-step runs, and gate_err signals that $C(t)$ is qualitative rather than quantitative.

A striking result is the 10×10 run at $\chi = 20$: despite gate_err ≈ 0.1 and large Bethe norm fluctuations, $C(t)$ follows the same qualitative trajectory as smaller grids (rapid decay to ≈ 0 by $t =$

Table 4: Finite-size scaling of Bethe approximation stability. $\chi = 50$ on MI100, except $n \times n \geq 8$ where $\chi = 20$. "Norm stable" means Bethe norm < 2.0 throughout the run. n^*: lightcone saturation layer (Table 2). gate_err at n^* is the accumulated truncation error at saturation.

Grid	χ	Steps	Norm stable	gate_err at n^*	$C(t)$ at n^*
3×3	50	20	✓	7×10^{-5}	0.111
4×4	50	20	✓	5×10^{-4}	0.056
5×5	50	30	✓	2×10^{-3}	0.034
6×6	50	30	✓	7×10^{-3}	0.033
7×7	60	40	✓	1×10^{-2}	0.002
8×8	20	40	✓†	6×10^{-2}	0.000
9×9	20	50	✓	$> 10^{-1}$	0.000
10×10	20	50	✓†	$> 10^{-1}$	0.000

† Norm exceeds 2.0 transiently but recovers; $C(t)$ is unaffected.

4.0), confirming that the scrambling physics is captured even when the variational approximation is far from exact. This demonstrates feasibility of the 10×10 geometry on MI100 at $\chi = 20$ and motivates the primary research target: 10×10 at $\chi = 100$ (requiring ≈ 109 GB, see §**??**).

1.6 Cross-Architecture Portability: Physics Comparison

The MI60 extended runs (this work) provide the first systematic comparison of $C(t)$ across AMD architectures for $\chi \in \{40, 60, 110\}$ and grid sizes 5×5 through 7×7.

4×4: portable operating points. Table 5 compares $C(t)$ at $t = 2.0$ for both architectures.

Table 5: $C(t)$ at $t = 2.0$ on 4×4: MI100 vs. MI60, all χ. **Bold**: metastable BP trap. Daggers mark the two portable operating points where both architectures agree and are trap-free.

χ	MI100	MI60	$\lvert\Delta C(t)\rvert$
20	0.0646 (trap L16)	0.0631 (trap L17)	0.0015
40	0.0572†	0.0577†	0.0005
50	0.0558	**0.0073**	—
60	0.0551†	0.0546†	0.0005
70	**0.0072**	0.054 (est.)	—
80	0.0543	**0.0073**	—
100	**0.0071**	0.054 (est.)	—
110	0.0537	Tensile fail	—

† Portable operating points: trap-free on both architectures, $|\Delta C(t)| < 10^{-3}$.

The results exhibit perfect complementarity in the trap pattern: MI60 traps at $\chi \in \{50, 80\}$ while MI100 is clean; MI100 traps at $\chi \in \{70, 100\}$ while MI60 is clean. Neither architecture is strictly more reliable than the other; the trap patterns are fingerprints of the BP fixed-point landscape selected by the MFMA reduction tree (MI100) vs. scalar vector-ALU (MI60).

At $\chi = 40$ and $\chi = 60$, both architectures are trap-free and agree to within 5×10^{-4} — better than

0.1%. These are the *portable* bond dimensions for this system.

MI60 ceiling: Tensile and BP. At $\chi = 110$, the MI60 run fails for two compounding reasons. First, BP enters a deep trap at $t = 0.7$ ($C(t) = 8.7 \times 10^{-5}$, sweeps = 100), indicating that the BP fixed-point landscape at this bond dimension is already problematic. Second, at the GEMM dimensions arising at $\chi = 110$ ($m = \chi^3 \approx 1.3 \times 10^6$, $n = D\chi = 440$), GFX906 lacks pre-built Tensile kernels and the fallback runtime compilation fails. The MI60 ceiling is $\chi \lesssim 100$ for correctness and $\chi \lesssim 80$ for reliable BP convergence.

Grid scaling on MI60. For 5×5 through 7×7 at $\chi = 50$, the MI60 runs with corrected centers produce $C(t)$ trajectories that are physically consistent with the MI100 odd_fix results. Table 6 shows $C(t)$ at saturation.

Table 6: $C(t)$ at n^* (lightcone saturation layer) for odd grids, $\chi = 50$, MI100 vs. MI60 (both with corrected geometric center). Agreement confirms that grid-size physics is portable at $\chi = 50$.

Grid	n^*	Steps	MI100 $C(t)(n^*)$	MI60 $C(t)(n^*)$	$\lvert\Delta C(t)\rvert$
5×5	20	30	0.057	0.057	< 0.001
6×6	30	30	0.038	0.033	0.005
7×7	30	30	0.039	0.033	0.006

The 5×5 result is portable to within the noise floor. The 6×6 and 7×7 discrepancies (~ 0.005) grow with grid size, consistent with the larger number of BP update paths and the growing sensitivity of the fixed-point selection to floating-point order. At these grid sizes, the dominant source of uncertainty in $C(t)$ is the BP fixed-point selection, not the truncation error.

1.7 Grid Scaling (Timing)

[Keep existing timing table and figure verbatim.]

1.8 Memory Model Validation

[Keep existing memory model section verbatim, with one addition:]

The chi=110 run on MI100 was measured at 94% VRAM utilization (≈ 30.1 GB of 32 GB), confirming the analytic prediction of ≈ 30 GB (Table **??**). The remaining 6% overhead is ROCm driver and context allocation. This establishes $\chi_{\max} = 110$ as the confirmed ceiling on a 32 GB MI100 for a 4×4 grid; the analytic model predicts $\chi_{\max} = 108$ as the limit, with the 2-unit gap explained by driver overhead not in the model.

References

[1] E. H. Lieb and D. W. Robinson, *The finite group velocity of quantum spin systems*, Commun. Math. Phys. **28**, 251 (1972).